

Samarth Desai

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github.com/Sammm03 — Portfolio

Professional Summary

First Class Honours BSc Computer Science (Cyber Security & Networks) graduate with hands-on experience building production-grade security tooling. Engineered KeyWatch, a real-time keystroke anomaly detection system achieving **92% precision** / **89% recall** using Isolation Forest on 5,000+ samples. Proficient in Python, SIEM-relevant UEBA logic, threat detection, network security, and GDPR-compliant system design. Actively seeking Graduate SOC Analyst or Junior Cyber Security Analyst roles.

Technical Skills

Security: SOC Analysis, UEBA, Anomaly Detection, Malware Analysis, Vulnerability Management, MITRE ATT&CK, Incident Response, OWASP Top 10, Endpoint Security, Access Control, Behavioural Biometrics

Frameworks & Standards: NIST SP 800-61r2, ISO/IEC 27001, GDPR (Articles 5/25/32), Least-Privilege Design, Privacy-by-Design

Tools & Platforms: SIEM, Wireshark, SolarWinds NPM, Cisco Packet Tracer, VMware, Isolation Forest (scikit-learn), Dash, Plotly, pynput

Programming: Python, Java, Bash Scripting, MVC Architecture, RBAC Implementation

Networking: IPv4/IPv6 Dual-Stack, MPLS, Leased Lines, Cisco 2911/Catalyst 2960, Subnetting, ifconfig/ip/nmcli

DevOps/Other: Git (version control), Linux (Ubuntu), Windows, macOS, Cross-Platform Development

Experience

Shift Manager

April 2023 – Present

Blue Ginger, Harrow, London, UK

- Served as primary on-site escalation point for operational incidents; coordinated real-time cross-functional response across 8+ staff — directly mirrors SOC Tier 1 triage and escalation workflows.
- Led incident triage and resolution under pressure, demonstrating prioritisation and composure transferable to first-line SOC on-call handling.
- Onboarded and trained 5+ team members on compliance procedures and operational standards, analogous to IT access provisioning and security policy enforcement.
- Enforced process controls via financial reconciliation and stock audit trails, comparable to IT asset management and governance audit disciplines.

Front of House Team Member

January 2023 – March 2023

Busaba, London, UK

- Supported customer-facing operations in a high-volume restaurant environment, demonstrating communication, adaptability, and team coordination under pressure.
- Managed table service, order handling, and guest experience across busy service periods, building composure and efficiency transferable to fast-paced operational roles.
- Collaborated closely with kitchen and management teams to resolve service issues in real time, reflecting cross-functional coordination under time-sensitive conditions.

Crew Member

October 2022 – December 2022

McDonald's, North Finchley, London, UK

- Operated under strict standard operating procedures and food safety protocols, building early discipline in process compliance and quality control checks.
- Handled high-throughput, time-sensitive tasks across a fast-paced environment, developing accuracy under pressure and operational speed.
- Worked within a structured team hierarchy, following escalation procedures for equipment faults and service issues to the correct point of contact.

Projects

KeyWatch — Keystroke Behaviour Monitoring & Anomaly Detection Dashboard

Jan 2025 – May 2025

University of Hertfordshire — Python, scikit-learn, Isolation Forest, Dash, Plotly, pynput

- Engineered real-time keystroke anomaly detection achieving **92% precision** / **89% recall** on 5,000+ labelled samples using Isolation Forest — directly deployable in UEBA/SIEM threat detection pipelines.
- Reduced dashboard I/O latency by **~40%** via MinMaxScaler normalisation and a hash-check plus 10-minute rolling-window retraining pipeline.

- Maintained dashboard latency consistently **below 12 seconds** on consumer hardware (avg CPU 33.35%), demonstrating production-readiness on resource-constrained environments.
- Delivered **GDPR Articles 5/25/32** compliance: local-only storage, anonymised timestamps, least-privilege file permissions — privacy-by-design from the ground up.
- Built interactive analyst dashboard with timeline graphs, anomaly visualisation, event logs, and Low/Medium/High sensitivity controls supporting continuous security monitoring.
- Architected modular, cross-platform codebase (Windows/macOS/Linux) separating keylogger, dashboard, and resource monitor for clean deployment and maintainability.

Parcel Depot Management System — Java Application

Oct 2024 – Dec 2024

University of Hertfordshire — Java, Swing, MVC, RBAC

- Built production-structured Java application with **RBAC across 5 class roles**, enforcing least-privilege access aligned with ISO 27001 access management principles.
- Implemented automated audit trail generation (fees, queue throughput, pending parcels) — mirrors SIEM and security logging requirements.
- Maintained 31 disciplined Git commits across the full project lifecycle demonstrating professional version-control workflow.
- Applied MVC architecture (controller, model, view packages) with Java Swing GUI for clear separation of concerns and operator usability.

Multi-Site Network Design & Linux Configuration

Sep 2024 – Dec 2024

University of Hertfordshire — Cisco Packet Tracer, Ubuntu/VMware, ifconfig, ip, nmcli

- Designed two-site enterprise network (London & Manchester) using **Cisco 2911 routers and Catalyst 2960 switches** with justified hardware selection against performance, scalability, and security criteria.
- Engineered **IPv4/IPv6 dual-stack** addressing with 27 subnets and /64 IPv6 prefixes per site; evaluated MPLS, leased lines, and broadband WAN options against real business requirements.
- Configured Ubuntu in VMware with role-segmented user accounts, least-privilege Linux permissions, and documented network setup using ifconfig, ip, and nmcli.
- Applied access control segmentation and minimised attack surface throughout architecture with documented security-first design decisions.

Education

University of Hertfordshire

September 2022 – May 2025

BSc (Hons) Computer Science — Cyber Security & Networks, **First Class Honours**

Key Competencies for Security Engineering Roles

- Threat Detection & Anomaly Analysis — Real-Time Incident Response — UEBA System Design
- Security-by-Design Architecture — Audit Trail Implementation — Policy & Compliance Enforcement
- Cross-Platform Systems Development — Network Security Architecture — Blue Team Operations